

Exercises: Dynamic Programming

Lecture 2/4, Reinforcement Learning

CentraleSupélec

Written Exercises: Pen and Paper

Topics covered

- **Bellman equations:** writing $v_\pi(s)$ and $v_*(s)$ from a given MDP.
- **Policy evaluation:** iterating the update $v(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)[r + \gamma v(s')]$ until convergence.
- **Policy improvement:** computing the greedy policy $\pi'(s) = \arg \max_a \sum_{s',r} p(s',r|s,a)[r + \gamma v(s')]$.
- **Policy iteration:** evaluation $\rightarrow$ improvement cycles until stability.
- **Value iteration:** iterating the Bellman optimality equation $v(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)[r + \gamma v(s')]$.

Structure

Today's exercises are organized as follows:

- Exercise 1 is a warm-up on action values, on the 4×4 gridworld of the lecture.
- Exercise 2 and 3 let you use Policy Improvement and Value Iteration on a small 3×3 gridworld to understand the mechanics of these algorithms and keep their definitions in mind.
- Exercise 4 is an introduction to the Gambler's problem, to do at home.
- The remaining exercises are reflections on the theory, you can do them at home if needed.
- The **Bonus: Proofs** section at the end (B1, B2) is optional.

Exercise 1: Warm-up on action values (Sutton & Barto's own gridworld, classic, Ch. 4, Example 4.1)

Recall the *real* gridworld of Example 4.1: 4×4 , states $0, \dots, 15$ numbered row by row, states 0 and 15 terminal, $r = -1$ per step, equiprobable random policy π_0 . Its converged v^{π_0} is already given in lecture (slide "Policy Evaluation — Value Function Evolution"):

$$v^{\pi_0} = [0, -14, -20, -22, -14, -18, -20, -20, -20, -20, -18, -14, -22, -20, -14, 0]$$

We can use the following relation:

$$q^{\pi_0}(s, a) = \sum_{s',r} p(s',r|s,a)[r + \gamma v^{\pi_0}(s')]$$

- (a) Using this table, compute $q^{\pi_0}(11, \text{down})$.
- (b) Compute $q^{\pi_0}(7, \text{down})$.
- (c) In one sentence, why is $q^{\pi_0}(11, \text{down})$ so much less negative than $q^{\pi_0}(7, \text{down})$, even though both take the same action?

Exercise 2: Policy improvement theorem on a 3×3 gridworld

Consider a 3×3 gridworld with:

- States: $S = \{1, 2, \dots, 8\}$ (numbered row by row), plus a terminal state **T** top-left.
- Actions: $A = \{\text{up, down, left, right}\}$, deterministic, actions that would leave the grid keep the agent in place.
- Reward: $r(s, a, s') = -1$ for every s, a, s' (except the terminal state).
- Discount factor: $\gamma = 1$ (episodic task).

Part A. Policy evaluation

Compute $v^{\pi_0}(s)$ for all states, where π_0 is the **equiprobable random** policy ($\pi_0(a|s) = 0.25$ for all a, s).

- Write the Bellman equation for $v^{\pi_0}(s)$.
- By symmetry, states group together. Exploit this symmetry.
- Solve the resulting linear system.

Part B. Action-value function (q^{π_0})

Compute $q^{\pi_0}(s, a)$ for each state s and each action a (at least for a few key states).

Same relation as in Exercise 1: $q^{\pi_0}(s, a) = \sum_{s', r} p(s', r|s, a)[r + \gamma v^{\pi_0}(s')]$

Part C. Testing the improvement theorem

The **policy improvement theorem** says that if you greedify the policy with respect to v^{π_0} :

$$\pi'(s) = \arg \max_a q^{\pi_0}(s, a)$$

then $v^{\pi'}(s) \geq v^{\pi_0}(s)$ for all s .

- Compute the greedy policy π' .
- Compare it with π_0 . Are they different?
- With no computation, evaluate $v^{\pi'}$ to check that $v^{\pi'}(s) \geq v^{\pi_0}(s)$.

Exercise 3: Value iteration on the 3×3 grid

We reuse exactly the environment of Exercise 2 (states $\{1, \dots, 8\}$ numbered row by row, terminal **T** top-left, $r = -1$ per step, $\gamma = 1$), but now solve the optimality equation directly instead of going through evaluation + improvement.

Start from $V_0(s) = 0$ for every non-terminal state.

- (a) Compute $V_1(s)$ for all states (a single sweep, $V_1(s) = \max_a [-1 + V_0(s')]$). What do you notice: are all 8 values equal? Why doesn't the grid's structure matter yet at this stage?
- (b) Compute $V_2(s)$ for all states. Comment what appears.
- (c) *Without computing further:* what will $V_k(s)$ converge to as $k \rightarrow \infty$?
- (d) Deduce an optimal policy π_* for each state. At least two states admit **several** tied optimal actions: which ones, and why? Connect this to the lecture's remark on the non-uniqueness of π_* (slide "Example 3.8").

Exercise 4: Value iteration: gambler's problem (reduced version), to do at home

A gambler bets on a sequence of **fair** coin flips ($p_h = 0.5$). They start with capital $s \in \{1, 2, 3\}$ and win as soon as they reach 4, and lose everything if they drop to 0. On each flip they stake an integer amount $a \in \{1, \dots, \min(s, 4 - s)\}$: heads $\Rightarrow$ capital $s + a$, tails $\Rightarrow$ capital $s - a$. Reward is zero everywhere except +1 on reaching 4. $\gamma = 1$.

- (a) Write the Bellman optimality equation for $v_*(s)$, $s \in \{1, 2, 3\}$ (set $v_*(0) = 0$, $v_*(4) = 0$; the +1 reward is already counted on the transition that reaches 4).
- (b) Check that $v(s) = s/4$ solves the Bellman equation for **any** policy (optimal or not), that is, for **every** possible stake a at each state.
- (c) What does this imply about the uniqueness of the optimal policy when $p_h = 0.5$? How many distinct optimal policies exist in that case?
- (d) (No computation) How does $v_*(s)$ qualitatively deform when $p_h < 0.5$? when $p_h > 0.5$? (cf. Sutton & Barto, example 4.3, for the sawtooth curves obtained when $p_h \neq 0.5$.)

Bonus: Proofs

Optional: two proofs reconstructing the theory behind the algorithms.

Bonus B1: Proof of the improvement theorem (theory bonus)

Formally prove the **policy improvement theorem**:

If $q^\pi(s, \pi'(s)) \geq v^\pi(s)$ for all $s \in S$, then $v^{\pi'}(s) \geq v^\pi(s)$ for all s .

Hint: Start from $v^\pi(s) \leq q^\pi(s, \pi'(s)) = \mathbb{E}_{\pi'}[R_{t+1} + \gamma v^\pi(S_{t+1}) \mid S_t = s]$, then use the hypothesis again at the next state: a recursion appears.

Bonus B2: Existence of a deterministic optimal policy (theory bonus)

The lecture stated (slide "Theorem: Optimal Policy Existence") that every finite MDP with bounded rewards has a deterministic optimal policy π_* , that $\pi_*(s) = \arg \max_a q_*(s, a)$, and that v_* is unique even though π_* may not be.

Assumption: $0 \leq \gamma < 1$. (For $\gamma = 1$, as in the episodic gridworlds above, the constant in Part A is 1: T^* is no longer a contraction and Banach's theorem does not apply directly.)

Part A: the Bellman optimality operator is a contraction. Define $T^* : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$ by

$$(T^*v)(s) = \max_a \sum_{s', r} p(s', r \mid s, a) [r + \gamma v(s')]$$

This is exactly one sweep of the value iteration you ran in Exercise 3.

- (a) For fixed s, a , show

$$\left| \sum_{s', r} p(s', r|s, a) \gamma [v_1(s') - v_2(s')] \right| \leq \gamma \|v_1 - v_2\|_\infty, \quad \|v\|_\infty := \max_s |v(s)|$$

- (b) Using $|\max f - \max g| \leq \max |f - g|$ (for any functions f, g of the action), deduce $\|T^*v_1 - T^*v_2\|_\infty \leq \gamma \|v_1 - v_2\|_\infty$: T^* is a γ -contraction.

Part B: existence and uniqueness of v_* . Use the Banach fixed-point theorem: a contraction on a complete space has a unique fixed point, reached from *any* starting point by iterating the map.

- (c) Explain why (b) + Banach guarantees that your value-iteration sweeps $V_{k+1} = T^*V_k$ converge to a unique v_* regardless of V_0 , and why v_* solves $v_* = T^*v_*$ (the Bellman optimality equation).
(d) Where exactly did you use finiteness of $\mathcal{S}$? Where did you use boundedness of the rewards?

Part C: the greedy policy attains v_* . Let $\pi_*(s) \in \arg \max_a \sum_{s', r} p(s', r|s, a) [r + \gamma v_*(s')]$ (finiteness of $\mathcal{A}(s)$ guarantees the $\arg \max$ is attained: this is the only place finiteness of the action set is needed).

- (e) Show that, by construction, $v_*(s) = \sum_{s', r} p(s', r|s, \pi_*(s)) [r + \gamma v_*(s')]$ for every s , i.e. v_* already solves the *policy evaluation* equation for π_* .
(f) The policy evaluation operator $T^{\pi_*}v(s) := \sum_{s', r} p(s', r|s, \pi_*(s)) [r + \gamma v(s')]$ is also a γ -contraction (same proof as Part A(a), without the $\max$). Its unique fixed point is $v_{\pi_*}()$ by definition. Combine this with (e) to conclude $v_{\pi_*}() = v_*$.

Part D: π_* beats every policy, and non-uniqueness.

- (g) Show that T^π is *monotone*: $v \leq v' \Rightarrow T^\pi v \leq T^\pi v'$ (pointwise), for any policy π . Use it to show $T^\pi v_* \leq T^*v_* = v_*$ for every π , then apply T^π repeatedly to both sides to conclude $v_\pi() \leq v_*$ for every π , so π_* (Part C) is indeed optimal.
(h) Suppose two actions $a_1 \neq a_2$ tie for the $\arg \max$ at some state s in the definition of π_* . Explain why both choices give a deterministic optimal policy, yet v_* itself does not change. Relate this to the ties you found in Exercise 3(d).

Notation & formulae

- **Policy evaluation** (fixed π , iterate to convergence):

$$v(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma v(s')] \quad \forall s$$

- **Policy improvement** (greedy w.r.t. v):

$$\pi'(s) \leftarrow \arg \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma v(s')]$$

- **Policy improvement theorem:** if $q^\pi(s, \pi'(s)) \geq v^\pi(s)$ for all s , then $v^{\pi'}(s) \geq v^\pi(s)$ for all s .
- **Policy iteration:** repeat evaluation $\rightarrow$ improvement until $\pi' = \pi$ (guaranteed to converge to π_* on a finite MDP).
- **Value iteration** (fold evaluation and improvement into one update):

$$v(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma v(s')] \quad \forall s$$

then extract $\pi_*(s) = \arg \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma v_*(s')]$ once v has converged.

- **Generalized Policy Iteration (GPI):** any interleaving of (possibly partial) evaluation and improvement steps converges to (v_*, π_*) : policy iteration and value iteration are the two extremes of the same idea.
- **Stopping criterion:** stop sweeping once $\max_s |v_{k+1}(s) - v_k(s)| < \theta$ for a small θ .